COUNTING FUNCTIONS IN THE COMPLETED NEVANLINNA CLASS

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AI assistance was used in adapting and checking the argument and the exposition.

Andersson asks: "Let $f \in \mathscr{N}$. Does there exist a set $E \subseteq \mathbb{T}^\infty$ of full measure such that if $\chi \in E$, then f_χ has an analytic continuation to $\mathbb{C}_0$ for which

$$\lim_{\sigma_0 \downarrow 0} \lim_{T \to \infty} \frac{\pi}{T} \sum_{\substack{s \in \mathscr{Z}(f_\chi - \zeta) \\ |\operatorname{Im} s| < T \\ \operatorname{Re} s > \sigma_0}} (\operatorname{Re} s - \sigma_0)$$

exists and is finite for all $\zeta \in \mathbb{C} \setminus \{f(+\infty)\}$?" Question 3, §6, pp. 26–30; zeros are counted with multiplicity.

Here $\mathscr{N}$ is the completion of uniformly convergent Nevanlinna Dirichlet series for the logarithmic metric $\|f - g\|_0 = \limsup_{\sigma \downarrow 0} \lim_{T \to \infty} (2T)^{-1} \int_{-T}^{T} \log(1 + |f - g|(\sigma + it))\, \mathrm{d}t$, and f_χ denotes the character twist, with $f(+\infty)$ its constant coefficient.

The source is Andersson, *The Littlewood–Paley formula and mean counting function for vertical limits of Dirichlet series*, Question 3 and Lemma 3.1, arXiv:2606.20293v2; that lemma is the extended nonnegative ergodic identity for the prime Kronecker flow.

Answered: yes. Evidence: Sketch.

Conditional on: arXiv:2606.20293v2, Lemma 3.1.

Yes. On one full-measure set the value is

$$J_\zeta(0+) - \log|f(+\infty) - \zeta|, \qquad J_\zeta(x) = \int_{\mathbb{T}^\infty} \log|f_\chi(x) - \zeta|\, \mathrm{d}m(\chi).$$

Proof. Write $d_0(q, r) = \|q - r\|_0$ for the metric defining $\mathscr{N}$, and let $\theta_t \chi$ denote the prime Kronecker flow. For $q \in \mathscr{N}_u$, the Bohr mean identity gives

$$\int_{\mathbb{T}^\infty} \log(1 + |q_\chi(s)|)\, \mathrm{d}m(\chi) \le \|q\|_0.$$

The disk submean inequality for the nonnegative subharmonic function $\log(1 + |q_\chi|)$ then gives, for every compact $K \Subset \mathbb{C}_0$,

$$\int \sup_{s \in K} \log(1 + |q_\chi(s)|)\, \mathrm{d}m(\chi) \le C_K \|q\|_0.$$

Apply this estimate to differences of a rapidly Cauchy sequence $q_j \in \mathscr{N}_u$ converging to f. Tonelli's theorem on a countable compact exhaustion shows that, almost surely, $q_{j,\chi}$ converges locally uniformly to an analytic representative F_χ of f_χ.

We also need control at infinity. Fix $u > 0$. If $x - u > 1$, the Bohr lift of $q(u + \cdot)$ and its analytic Poisson evaluation give

$$\sup_{\chi, t} \log(1 + |q_\chi(x + it)|) \le \prod_p \frac{1 + p^{-(x-u)}}{1 - p^{-(x-u)}}\, \|q\|_0.$$

The product is finite. Applying this to $q_j - q_k$ shows that the approximants are uniformly Cauchy on a sufficiently far-right half-plane, uniformly in χ and t. Their constant coefficients are Cauchy as well, since $\log(1 + |a_1(q_j - q_k)|) \le d_0(q_j, q_k)$. Since each fixed q_j tends to its constant coefficient there, we obtain the uniform anchor

$$\lim_{x \to \infty} \sup_{\chi, t} |F_\chi(x + it) - a_1| = 0, \qquad a_1 = f(+\infty).$$

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In particular, for any compact target set $A \Subset \mathbb{C} \setminus \{a_1\}$, all F_χ avoid A to the right of one common vertical line.

Fix such an A and a positive cutoff σ_*. Jensen's disk inequality and the compact-sup estimate above give an integrable random bound for the number of ζ-points, uniformly over $\zeta \in A$, in any fixed-width rectangle contained in $\{\operatorname{Re} s \geq \sigma_*\}$. For a continuous compactly supported weight h in that half-plane, the divisor sum $H_\zeta(\chi) = \sum_{F_\chi(s)=\zeta} h(s)$ is continuous in ζ and has this integrable envelope. Apply the ergodic theorem for θ_t to finite target nets and to the moduli of continuity of H_ζ; this yields uniform convergence of the time averages over $\zeta \in A$. Choose h to smooth the vertical cutoff and the factor $(\operatorname{Re} s - \sigma_0)_+$. The smoothed average differs from the sharp count only in bounded-width blocks at the two vertical ends. The integrable Jensen bound and Borel–Cantelli make that error $o(1)$ after division by T, uniformly on A. Taking countable compact exhaustions of $\mathbb{C} \setminus \{a_1\}$ and rational $\sigma_0 > 0$ gives one full-measure set where the inner limit exists for every such target and cutoff. Rational cutoffs on either side and continuity extend the assertion to every real $\sigma_0 > 0$.

To identify the value, fix $\zeta \neq a_1$. Jensen's inequality for the Dirichlet polynomials, followed by approximation in d_0, gives $J_\zeta(x) \geq \log|a_1 - \zeta|$; the gauge bound gives a finite upper bound. Thus J_ζ is finite and convex. Poincaré–Lelong's formula, averaged over characters, says that the expected zero measure per unit vertical length is $J_\zeta''(dx)/(2\pi)$. The uniform anchor implies that F_χ has no ζ-points beyond a common far-right line and that $J_\zeta(x) \to \log|a_1 - \zeta|$ there; convexity then makes J_ζ constant past that line. Therefore integration by parts gives, for each $\sigma_0 > 0$,

$$2\pi \int_{x>\sigma_0} (x - \sigma_0) \frac{J_\zeta''(dx)}{2\pi} = J_\zeta(\sigma_0) - \log|a_1 - \zeta|.$$

The ergodic limit of the sharp counts equals this expectation. Finally, the finite upper bound for J_ζ makes $J_\zeta(0+)$ finite; letting $\sigma_0 \downarrow 0$ proves the stated formula in the required order of limits. If f is constant, all counts for $\zeta \neq a_1$ vanish, so the claim is immediate. □